

Cite this: *Phys. Chem. Chem. Phys.*, 2012, **14**, 5147–5152

www.rsc.org/pccp

PAPER

Ionic liquid nanotribology: mica–silica interactions in ethylammonium nitrate

Oliver Werzer,^a Emily D. Cranston,^{†b} Gregory G. Warr,^c Rob Atkin^{*a} and Mark W. Rutland^{*bd}

Received 4th October 2011, Accepted 1st November 2011

DOI: 10.1039/c1cp23134k

Colloid probe atomic force microscopy has been used to study the nanotribological properties of the silica–ethylammonium nitrate (EAN)–mica system. Normal force curve measurements reveal a series of steps at separations that are consistent with the size of an EAN ion pair (0.5 nm) due to displacement of structured solvent layers as the two surfaces are brought together. At closest separations, two steps are measured with widths of 0.3 nm and 0.1 nm, which are too small to be due to an ion pair layer. The 0.3 nm step is attributed to a partial displacement of a silica-bound cation-rich layer, with residual cations being removed in the subsequent 0.1 nm step. Lateral force measurements reveal that the frictional response is dependent on the number of ion pair layers between the surfaces. At low forces, when there is more than a single layer of EAN between silica and mica, the lateral force increases relatively steeply with applied load, and is independent of the sliding speed. At intermediate forces, a single layer of cations in an intercalated bilayer structure is present between the surfaces. The friction coefficient (μ) increases logarithmically with sliding speed consistent with an activated, discontinuous sliding process. At high force, μ is small and once again, independent of sliding velocity. The adsorbed cation layer is bound primarily to mica and compressed by the high normal force. This robust layering with a well-defined sliding plane permits the colloid probe to slide easily over the mica surface.

Introduction

Ionic liquids (ILs) are molten salts that have melting points below 100 °C. Low melting points are achieved *via* the use of bulky, sterically hindered ions that prevent packing into a crystal lattice.¹ IL physical properties can be varied in a controlled fashion through systematic changes in the molecular structure of the ions, which has led to them being referred to as ‘tunable’ or ‘designer’ solvents. This tunability, and the unusual set of physical properties typically exhibited by ILs (which includes low vapour pressure, high temperature stability, and high thermal and electrical conductivity), has led to research interest in a diverse range of fields including particle stabilization,² synthesis,¹ catalysis,³ CO₂ capture,⁴ solar cells,⁵ organic field effect transistors,⁶ batteries,⁷ extractions,⁸ as solvents for surfactant self-assembly,^{9–13} proteins,^{14–16} and

polymers,^{17–20} solid–liquid^{21–24} and liquid–air^{25,26} interfacial phenomena, amongst many others.

ILs’ potential as lubricants was recognised more than a decade ago.²⁷ Their thermal conductivity and stability mean they are able to withstand high pressures and readily dissipate heat, which are desirable characteristics for a lubricant; a significant number of papers have examined thus IL lubrication macroscopically.^{27,28}

More recently, experiments have revealed that the nanoscale properties of ILs may also be advantageous for lubrication. Scattering experiments have shown that many ILs are nanostructured in the bulk,^{29,30} and surface force measurements have shown that this structure manifests itself as layering when ILs are confined between solid surfaces.^{31,32} As both the liquid nanostructure^{30,33,34} and the strength with which the IL interacts with solid surfaces^{22,23} can be tuned, it should be possible to optimize ILs for lubrication at a molecular level. Compared to molecular lubricants, the IL ions interact strongly with oppositely charged surfaces and resist ‘squeezing out’ as the contact is compressed, meaning that a lubricating film will remain in place up to higher applied normal forces.³⁵ As most ILs also have apolar moieties, and thus amphiphilic character, similar attractive interactions can occur with uncharged surfaces.³¹ However, only a handful of papers have examined IL nanotribology^{27,35} and attempted to elucidate the nanoscale factors that control IL

^a Discipline of Chemistry, The University of Newcastle, Callaghan, NSW, 2308, Australia. E-mail: rob.atkin@newcastle.edu.au

^b Surface and Corrosion Science, Department of Chemistry, Royal Institute of Technology, SE-100 44 Stockholm, Sweden. E-mail: mark@kth.se

^c School of Chemistry, The University of Sydney, NSW, 2006, Australia

^d Institute for Surface Chemistry, Stockholm, Sweden

[†] Current address: Chemical Engineering, McMaster University, Hamilton, ON, Canada.

lubrication. This paper probes this issue, through correlation of the normal and lateral force data for ethylammonium nitrate (EAN) confined between mica and silica surfaces.

EAN is a protic IL,³⁶ synthesized by proton transfer from nitric acid to ethylamine resulting in ethylammonium cations (EA^+) and nitrate anions. EAN has a melting point of 12 °C, density of 1.2 g mL⁻¹ and is 30 times more viscous than water. Both ions have three H-bonding sites which results in the formation of an extended H-bonded network.³⁷ Neutron scattering experiments have revealed that bulk EAN has a sponge-like nanostructure,^{29,33} which results from electrostatic and hydrogen bonding attractions between the ethylammonium cation and the nitrate anion that favors the formation of ionic domains. This produces a solvophobic force³⁸ that causes the cation's alkyl groups to aggregate into nonpolar domains. The amphiphilic character of EAN is central to explaining its structure and properties; in the vicinity of a macroscopic interface, EAN order becomes even more pronounced,^{31,32} similar to the "sponge to lamellar" transition that occurs for aqueous surfactant phases near a surface³⁹ or under confinement.⁴⁰ Interfacial EAN layers are readily detected by measurement of the force profile normal to the interface using an atomic force microscope^{22,31} (AFM) or surface forces apparatus³² (SFA).

Nanotribological measurements are performed by compressing two surfaces together at systematically varied normal loads (F_{norm}) and measuring the force of lateral resistance (F_{lat}) as the surfaces are sheared. Such measurements have earlier proven highly useful for evaluating for example friction mechanisms of interfacial films⁴¹ and how pre-contact surface forces can be used to mediate friction.⁴² In this work we use AFM to investigate how F_{lat} varies with F_{norm} for the silica–EAN–mica system, to reveal how friction varies with the number of EAN layers between the surfaces.

Materials and methods

The structure of ethylammonium nitrate (EAN) is shown in Fig. 2. It was prepared by reacting equimolar amounts of ethylamine (Sigma Aldrich, Munich, Germany) and concentrated nitric acid (Merck, Darmstadt, Germany) in excess Milli-Q water (Millipore, Malsheim, France). The solution temperature was maintained at 8–10 °C to prevent the formation of oxide impurities. Water was removed by rotary evaporation at 50 °C, then purging with nitrogen and heating at 105–110 °C for 12 h. This method resulted in H₂O contents undetectable by Karl-Fischer titration.

Normal and lateral force measurements were obtained using a Nanoscope IIIa Multimode AFM (Veeco[®], Santa Barbara, CA)

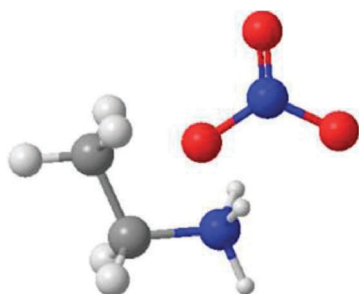


Fig. 1 Molecular structure of EAN.

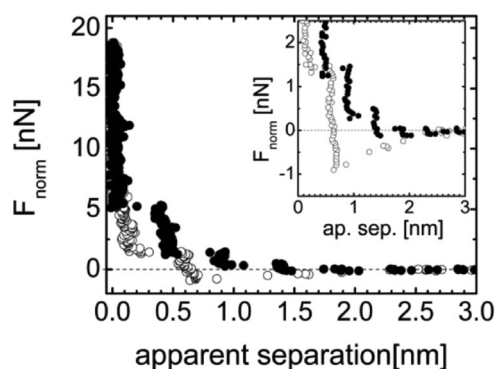


Fig. 2 Normal force curve measurement of a 20.8 μm silica colloid probe approaching (closed circles) and retracting (open circles) from the EAN–mica interface. The inset shows similar data on an expanded vertical scale to highlight the features at low forces.

equipped with a PicoForce controller and PicoForce scanner in contact mode. A silica particle (Bangs Laboratories, Fishers, IN) was attached to a tipless rectangular cantilever (model CSC12, Mikromasch, Tallinn, Estonia) using Araldite[™] epoxy under an optical microscope. The colloid particle radius was determined to be 10.4 μm. The normal ($k_z = 0.98 \text{ N m}^{-1}$) and torsional ($k_t = 8.74 \times 10^{-9} \text{ Nm/rad}$) spring constants were determined by the thermal vibration method following the procedure of Sader.⁴³

Mica surfaces were freshly cleaved prior to the experiment. AFM experiments were carried out in a liquid cell sealed with a silicon rubber O-ring that prevents water ingress. The liquid cell and the O-ring were cleaned by ultrasonication in 0.01 wt% cetyltrimethylammonium bromide solution, rinsing with ethanol (Kemetyl, Haninge, Sweden) and Milli-Q water and finally dried with nitrogen. Normal force curve measurements and friction force measurements were performed after introducing 1 mL EAN into the cell and equilibrating for at least 30 min.

AFM normal force curve measurements were performed by moving the surface towards the colloid probe and recording the cantilever deflection as a function of separation. To see structuring in EAN, force curves were collected over 50 nm at scan rates of 4–10 nm s⁻¹; the surfaces were kept in contact for 1 s before separating which allowed for "structural steps" to be observed on retraction as well. Deflection vs. separation curves were converted to force vs. apparent separation curves using standard methods.⁴⁴ By convention, the region of constant compliance is used to define the zero separation in an AFM experiment. However, it is possible that strongly adsorbed species remain between the probe and the surface even at high force. To account for this, the quoted distance, ostensibly between the tip and the surface, is referred to as the "apparent separation".

Friction measurements were obtained by performing lateral AFM scans with a scan angle of 90° (with respect to the cantilever long axis) with the fast scan axis disabled. The lateral deflection signal (*i.e.*, cantilever twist) was converted to lateral force using the FrictionIT software (ForceIT, Sweden) which takes into account the torsional spring constant and the geometrical dimensions of the cantilever. The lateral deflection sensitivity of the AFM's photodiode detector was calibrated according to the torsional method.⁴⁵ Friction force data were

obtained for different speeds and normal force loads. Each individual lateral force data point is an average of at least six repeat friction loops to produce statistically significant data.

Results and discussion

Normal forces

Force data for the approach and retraction of a 20.8 μm diameter silica colloid probe interacting with a mica substrate in the presence of EAN are presented in Fig. 2. These data are in accordance with that described previously for similar systems.^{22,23,31} A series of steps are observed as the major feature, which are a consequence of interfacial solvent layering.³¹ The force required to push through each EAN layer increases at smaller separations, indicating that the ordering is stronger closer to the interface. Steps are also seen as EAN layers re-order upon retraction of the probe from the substrate since any significant adhesion which might be associated with a Lifshitz interaction is absent. Also absent is the longer range double layer force observed in water and which might be expected in a polar solvent. In both cases, the lack of observed interaction is due to the molten salt nature of the EAN solvent, with an effective ionic strength of 11 M. Since both the Lifshitz and double layer interaction have an electromagnetic origin, this ionic strength is more than sufficient to completely screen the interaction.

At large separations (in this case >3 nm) no force is measured as the probe moves through the bulk liquid towards the substrate. At a separation of 2.4 nm a weak interfacial layer is encountered which resists approach until a force of ~ 0.1 nN is reached (see inset to Fig. 2), whereupon the layer is ruptured and the tip jumps inward by 0.5 nm (consistent with the molecular dimensions of an EAN ion pair³²) to the next solvent layer. This process is repeated four times, with the force required to rupture each subsequent layer increasing, until an apparent separation of 0.4 nm is reached. The probe expels the layer at 0.4 nm when a force of 5 nN is reached, and jumps to a separation of only ~ 0.1 nm. This final barrier gives way at a force of 13 nN. This small step distance is close to the spatial resolution of the instrument, but it will be shown below that a marked change in the friction behaviour also occurs at a force of 13 nN, tending to confirm the existence of this feature. As the widths of the two steps at smallest separations (0.3 nm and 0.1 nm) are less than the EAN ion pair dimension, they must be due to displacement of either an individual cation or anion layer, or to the rearrangement of the ions within a layer. Prior work using sharpened silica nitride AFM tips and various surfaces has shown that EA^+ remains adsorbed to the mica surface up to at least the maximum force that can be applied with the AFM, but that the cations can be displaced from the silica surface.³¹ In that work, displacement of the cation layer from the silica substrate resulted in a step width of 0.3 nm, so it is reasonable to suggest that the 0.3 nm step measured in the current work is similarly due to a cation-rich layer being displaced from the surface of the silica colloid probe. It is very difficult to prove irrefutably that it is from the silica surface that the layer is displaced. There are, however, a few arguments which can be marshalled to support this contention and a brief discussion of the adsorption and layering is now undertaken. In the molten salt, both surfaces are expected to be close to fully

charged (in the case of silica the majority of the silanol groups will be dissociated) and thus each surface is expected to be coated with an adsorbed layer consisting of predominantly EA^+ cations. (Whether the layer should be considered adsorbed or part of a lamellar ordering is a matter of semantics for this system.) The subsequent layer is also expected to consist predominantly of cations since the charge of the first layer is largely balanced by the charge on the silica/mica surface and solvophobic interactions are likely to dominate the interaction between the primary and secondary layers. For charge neutrality, each of these layers must have a layer of anions associated with it and these act as “ion layer pairs”. This surface-induced lamellar layer structure decays over a few layers to the bulk sponge phase. The reason it is not perfectly symmetrical (*i.e.*, the silica-bound layer can be displaced but not the mica-bound layer) can be explained in several ways. While silica⁴⁶ has a higher density of silanol groups than the charge density of mica⁴⁷ it is not clear whether *all* the silanol groups would be dissociated. Silica is also slightly rougher than mica so its capacity to order the liquid may be slightly less. Probably most importantly, the silanol sites on silica consist of a covalent bond between an oxygen atom and a proton. When an EA^+ cation is associated with a dissociated silanol site, a significant hydrogen bonding contribution between the adsorbate and the surface site would therefore be expected. When such an adsorbed layer is displaced, a proton may remain on the silanol group, rendering the silica surface neutral to preserve net charge neutrality, while neutral ethylamine is expelled into the bulk.

Irrespective of the exact mechanism, when this layer is removed, the silica and mica surfaces are separated by only the thickness of a single layer of cations, most of which are strongly bound to the mica, and it is this configuration which defines the zero of apparent separation in the AFM.

The precise origin of the 0.1 nm step cannot be deduced with certainty, but it is too small to correspond to displacement of even a single ion layer. Rather, it probably arises as a consequence of changes in the final (cation) layer structure. Since the size of the EA^+ cation is smaller than the average area per charge on the mica surface (around 0.5 nm^2),⁴⁷ there are likely to be additional cations in the layer with an associated anion for charge neutrality. It may be that on compression some pairs are expelled, allowing a structural rearrangement of the final layer. This rearrangement may take the form of a transition from a relatively upright position (with hydrocarbon moiety pointing outwards to facilitate solvophobic interactions with the next layer) to one where the alkyl chains are orientated more parallel to the surface. Alternatively, it may be that the cations adsorbed to the mica undergo a liquid-to-solid (ordered or amorphous) transition at high normal force, producing a small decrease in layer thickness. Like many ILs, EAN is known to be a good glass-former. Finally, it may result from incomplete cation desorption from the silica surface at lower loads in which case the step would thus be the result of removal of these residual species, which would necessarily be highly intercalated with the mica bound layer. However we discount surface roughness or porosity of the silica as possible explanations, since the final step size and push-through force was quite consistent between different experiments and colloid probes.

The subtlest features of the structure remain necessarily obscure but the important conclusions of a surface induced lamellar-like structure, with ion pairs being sequentially removed on approach is unambiguous.

Lateral forces

Fig. 3 shows the lateral (friction) force, F_{lat} acting on a 20.8 μm diameter silica colloid moving at various speeds over a distance of 5 μm across the EAN–mica interface as a function of normal applied load, F_{norm} . The corresponding approaching normal force curve (*cf.* Fig. 2) is shown on a common load abscissa. The lateral force increases with normal load over the full range of lateral speeds examined. The lateral force response can be decomposed into 2–3 linear regions (Interval I, II and III), each having a distinct slope. This behaviour is insensitive to system history, and the same behaviour is observed both on increasing and decreasing load. Repeat measurements under the same experimental conditions show very similar response, suggesting that the sample is not affected by wear during the course of an experiment. Small variations in the location of the interval boundaries were observed which, however, are comparable to the variations in the rupture forces seen in the normal force curves (*vide infra*).

As the applied normal force is increased from 0 nN, the apparent separation decreases from $>2\text{ nm}$ to 0.4 nm as four weak ion pair layers are ruptured, and then remains almost constant up to 5 nN load (Interval I). Over the same normal force range, the friction force increases linearly, and is independent of sliding speeds between 5 and 40 m s^{-1} . The friction coefficient, μ , for this interval is obtained from Amontons' law, $F_{\text{lat}} = \mu F_{\text{norm}}$ to be 0.14, and is similar to values obtained in previous IL AFM

lubrication studies.³⁵ At lower sliding speeds (0.5 to $1\text{ }\mu\text{m s}^{-1}$) the lateral forces and corresponding friction coefficient are much smaller.

The near-constant 0.4 nm apparent separation is less than the EAN ion pair dimension and, as noted above, the zero of apparent separation must correspond to the strongly adsorbed cation layer on mica. Hence this apparent separation likely corresponds to a layer of cations adsorbed to the mica surface and a layer of ions enriched in cations interacting with the silica substrate.

When the load exceeds 5 nN the apparent separation decreases to $\sim 0.1\text{ nm}$ (Interval II). Here, a break is seen in the friction data, which is most noticeable at intermediate sliding speeds. At this small separation only a single confined layer remains, consisting mostly of cations adsorbed to mica in order to neutralize its surface charge. Here the friction coefficient depends strongly on lateral velocity. This friction coefficient, defined by the gradient of the lateral *versus* normal force using a modified Amontons' law, $F_{\text{lat}} = \mu F_{\text{norm}} + F_{\text{cont}}$, is smaller at most sliding speeds than that obtained in Interval I for thicker confined films, at lower normal load. Interestingly, Perkin *et al.* found that μ increased for [Emim][EtSO₄] confined between two mica surfaces over an analogous structural transition.³⁵

When the normal load is further increased above 13 nN a final transition (to Interval III) is observed, which corresponds to a decrease in the apparent separation to zero. As noted above, this is interpreted as a structural transition in the mostly-cation layer adsorbed onto mica. Over this interval, the friction coefficient is further reduced and once again independent of lateral velocity.

The lateral forces measured for the slowest velocities ($0.5\text{ }\mu\text{m s}^{-1}$ and $1\text{ }\mu\text{m s}^{-1}$) are much smaller than those measured for $5\text{ }\mu\text{m s}^{-1}$ for all normal forces, and these measurements displayed greater variability. The frictional force measured for a probe speed $0.5\text{ }\mu\text{m s}^{-1}$ thus appears to be greater than that for $1\text{ }\mu\text{m s}^{-1}$ for normal forces greater than 13 nN , which seems unlikely. For this reason, the molecular origin of the lateral force response for the slowest colloid speeds is not addressed below.

When the tip velocity is $5\text{ }\mu\text{m s}^{-1}$ or greater and the normal force is low (Interval I), the lateral force is independent of the sliding speed *i.e.*, when the surfaces have more than a single layer of ions between them the lateral force depends only on the normal load, and not the colloid speed for velocities up to at least $40\text{ }\mu\text{m s}^{-1}$. This suggests that any rearrangements that occur within the adsorbed layers are fast at these low applied loads.

Fig. 4 shows how the friction coefficients depend on the lateral velocity of the colloid when only a single layer separates the silica and mica surfaces. In Interval II, μ increases logarithmically with colloid probe speed up to the maximum speed examined. This behavior has been observed previously for lubricated sliding of Langmuir–Blodgett films,⁴⁸ as well as confined molecular liquids,⁴⁹ in both SFA and AFM studies, and is explained by an activated, discontinuous sliding process.

In Interval II the single confined layer is not fully compressed, and for the reasons discussed in the preceding section is likely to have a larger roughness associated with it. (For example, it may contain EA^+ cations bound to both silica and mica charged

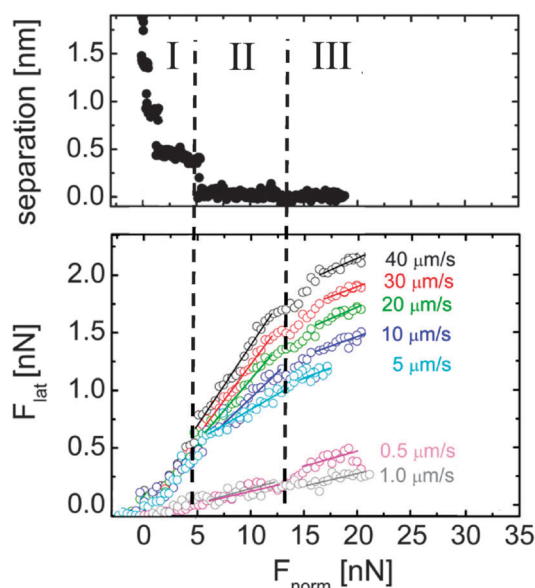


Fig. 3 (top) Separation *versus* normal force. This is the same data shown in Fig. 1, but with the axes reversed for easy comparison to the lateral force data. (bottom) Force measurements for various sliding velocities *versus* normal force. Dashed vertical lines denote the divisions between Intervals I, II, and III as labeled in the figure. Solid lines denote linear fits to increasing load data from which friction coefficients were obtained.

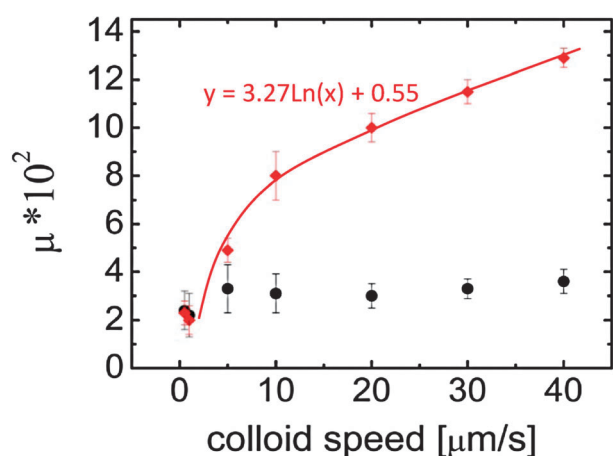


Fig. 4 Colloid probe speed dependent friction coefficient of Interval II (diamonds) and III (circles). The solid line shows a logarithmic line of best fit for Interval II.

sites as an asymmetric, highly-intercalated bilayer.) Irrespective of the exact nature of the transition from Interval II to III, a periodic structure would be expected in Interval II and sliding of the colloid probe across this penultimate surface layer would then result in a Tomlinson⁵⁰-like response. (A stick slip-like process at the “molecular” level – with harmonic potentials given by the local topography and the effective spring constants of the surface bound features). Such models, predicting the logarithmic dependence of friction on velocity, have successfully been used to model the “atomistic” friction of crystals under UHV,⁵¹ as well as the dynamic friction response of confined liquids of large dimensions⁴⁹ and polymers.⁵² The prefactor to the logarithm of velocity term in the expression relating friction force to velocity is related to the Tomlinson potential. In this work this value is of the order of 0.3 nN which is much less than the value (about 3 nN) for the considerably larger molecules of OMCTS⁴⁹ and much larger than for NaCl (0.03 nN). Given that the structural features expected to give rise to the frictional response are intermediate between these two extreme cases, this intermediate value must be considered reasonable.

The friction coefficient in Interval III is smaller and any dependence on sliding velocity is undetectable within experimental error. In this interval the adsorbed layer is compressed by the high normal force, leading to a smoother layer and thus allowing easy sliding of the probe across the cation-coated mica surface. If the 0.1 nm step in the normal response is in fact due to displacement of residual, silica-bound cations, then the confined film is essentially a mica-adsorbed monolayer. The silanol groups on the silica colloid surface no longer contain sticking sites for the confined film, and there are no more intercalated tails to brush past each other. Low friction would be expected as the silica probe slides across the alkyl-chain coating on the mica.

Here again the amphiphilic nature of the ethylammonium cation seems to play a central role in the behaviour of EAN, enabling the formation of highly-oriented “tail-out” adsorbed monolayers and intercalated bilayers. In contrast, [EMIm]⁺ is a more symmetrical ion, and the detailed interaction of the imidazolium ring with the surface leads to quite different lubrication behavior.³⁵

Conclusions

The nanotribology of structured EAN films between silica and mica surfaces has been investigated using normal and lateral colloid probe AFM force measurements. Correlation of the high resolution normal force data with the lateral forces allows the load to be indirectly used as a measure of separation and precontact surface forces can be used to explain the frictional variation with load. For all lateral probe velocities investigated, the lateral force increased with normal load, and three distinct regimes were observed, characterised by different friction coefficients. Comparison to normal force data revealed that these different friction coefficients are correlated directly with steps in the normal force curves, meaning that μ is dependent on the number and structure of ion layers between the surfaces.

Neither the velocity dependence, nor the magnitude of the friction coefficient, appear to be monotonic with the number of layers. At low loads where there are several ionic layers between the surfaces, the friction coefficient is apparently independent of velocity and is relatively high. At higher loads a transition occurs to a region containing a single, weakly compressed layer of ions. These cations form an asymmetric intercalated partial bilayer which is predominantly attached to the mica surface. As the surfaces are sheared, this structure results in an activated sliding process, and the friction coefficient is dependent on the sliding speed. For the highest rates its magnitude is comparable to that for multilayer separation. Displacement of residual cations from the silica surface results in a monolayer of apparently homogeneous structure which, due to its amphiphilic alignment, acts as a well-defined shear plane. Consequently the friction coefficient is low, and once again independent of the sliding speed.

The amphiphilicity controlled ordering characteristics observed under confinement and shear are consistent with other studies of EAN which have revealed that the amphiphilicity of the cation is a key determinant of nanostructure and physical properties. “Tail-out” adsorbed monolayers and intercalated bilayers have both been detected at the interface, and markedly different frictional responses measured for the two different structures.

From a nanotribological perspective ionic liquids are increasingly attracting interest due to their high thermal stability and enhanced viscosity. These studies show that they also act as intrinsic boundary lubricants, due to their capacity to bind strongly to surfaces and form lamellar structures.

Acknowledgements

This work was funded by the Australian Research Council (DP0986194). MR acknowledges support from the Swedish Research Council (VR) and the Foundation for Strategic Research, SSF. (Biomime™ and Microstructure for Friction and Corrosion Control). The AFM was provided by a generous grant from the Knut and Alice Wallenberg Foundation.

References

- 1 T. Welton, *Chem. Rev.*, 1999, **99**, 2071.
- 2 J. A. Smith, O. Werzer, G. B. Webber, G. G. Warr and R. Atkin, *J. Phys. Chem. Lett.*, 2010, **1**, 64.

- 3 T. J. Geldbach and P. J. Dyson, *J. Am. Chem. Soc.*, 2004, **126**, 8114.
- 4 H. Rodriguez and R. D. Rogers, *Fluid Phase Equilib.*, 2010, **294**, 7.
- 5 M. Gratzel, *Nature*, 2001, **414**, 338.
- 6 J. Lee, M. J. Panzer, Y. He, T. P. Lodge and C. D. Frisbie, *J. Am. Chem. Soc.*, 2007, **129**, 4532.
- 7 J.-W. Choi, G. Cheruvally, Y.-H. Kim, J.-K. Kim, J. Manuel, P. Raghavan, J.-H. Ahn, K.-W. Kim, J.-J. Ahn, D. S. Choi and C. E. Song, *Solid State Ionics*, 2007, **178**, 1235.
- 8 L. A. Blancard, D. Hancu, E. J. Beckman and J. F. Brennecke, *Nature*, 1999, **399**, 28.
- 9 M. U. Araos and G. G. Warr, *J. Phys. Chem. B*, 2005, **109**, 14275.
- 10 M. U. Araos and G. G. Warr, *Langmuir*, 2008, **24**, 9354.
- 11 R. Atkin and G. G. Warr, *J. Am. Chem. Soc.*, 2005, **127**, 11940.
- 12 R. Atkin, S. M. C. Bobillier and G. G. Warr, *J. Phys. Chem. B*, 2010, **114**(3), 1350.
- 13 T. L. Greaves and C. J. Drummond, *Chem. Soc. Rev.*, 2008, **37**, 1709.
- 14 N. Byrne and C. A. Angell, *Chem. Commun.*, 2009, 1046.
- 15 N. Byrne, L.-M. Wang, J.-P. Belieres and C. A. Angell, *Chem. Commun.*, 2007, 2714.
- 16 J. P. Mann, A. McCluskey and R. Atkin, *Green Chem.*, 2009, **11**, 785.
- 17 O. Werzer, G. G. Warr and R. Atkin, *Langmuir*, 2011, **27**, 3541.
- 18 O. Werzer, G. G. Warr and R. J. Atkin, *J. Phys. Chem. B*, 2011, **115**, 648.
- 19 Y. He, Z. Li, P. Simone and T. P. Lodge, *J. Am. Chem. Soc.*, 2006, **128**, 2745.
- 20 R. Atkin, L.-M. De Fina, U. Kiederling and G. G. Warr, *J. Phys. Chem. B*, 2009, **113**, 12201.
- 21 R. Atkin, S. Z. E. Abedin, R. Hayes, L. H. S. Gasparotto, N. Borisenko and F. Endres, *J. Phys. Chem. C*, 2009, **113**, 13266.
- 22 D. Wakeham, R. Hayes, G. G. Warr and R. Atkin, *J. Phys. Chem. B*, 2009, **113**, 5961.
- 23 R. Hayes, G. Warr and R. Atkin, *Phys. Chem. Chem. Phys.*, 2010, **12**, 1709.
- 24 F. Endres, O. Hoff, N. Borisenko, L. H. Gasparotto, A. Prowald, R. Al-Salman, T. Carstens, R. Atkin, A. Bund and S. Z. E. Abedin, *Phys. Chem. Chem. Phys.*, 2010, **12**, 1724.
- 25 D. Wakeham, P. Niga, G. G. Warr, M. W. Rutland and R. Atkin, *Langmuir*, 2010, **26**, 8313.
- 26 P. Niga, D. Wakeham, A. Nelson, G. G. Warr, M. W. Rutland and R. Atkin, *Langmuir*, 2010, **26**, 8282.
- 27 M. Palacio and B. Bhushan, *Tribol. Lett.*, 2010, **40**, 247.
- 28 I. Minami, *Molecules*, 2009, **14**, 2286.
- 29 R. Atkin and G. G. Warr, *J. Phys. Chem. B*, 2008, **112**, 4164.
- 30 A. Triolo, O. Russina, H. J. Bleif and E. DiCola, *J. Phys. Chem. B*, 2007, **111**, 4641.
- 31 R. Atkin and G. G. Warr, *J. Phys. Chem. C*, 2007, **111**, 5162.
- 32 R. G. Horn, D. F. Evans and B. W. Ninham, *J. Phys. Chem.*, 1988, **92**, 3531.
- 33 R. Hayes, S. Imberti, G. G. Warr and R. Atkin, *Phys. Chem. Chem. Phys.*, 2011, **13**, 13544.
- 34 R. Hayes, S. Imberti, G. G. Warr and R. Atkin, *Phys. Chem. Chem. Phys.*, 2011, **13**, 3237.
- 35 S. Perkin, T. Albrecht and J. Klein, *Phys. Chem. Chem. Phys.*, 2010, **12**, 1243.
- 36 P. Walden, *Bull. Acad. Imp. Sci.*, 1914, 1800.
- 37 K. Fumino, A. Wulf and R. Ludwig, *Angew. Chem., Int. Ed.*, 2009, **48**, 3184.
- 38 A. Ray, *Nature*, 1971, **231**, 313.
- 39 W. A. Hamilton, L. Porcar, P. D. Butler and G. G. Warr, *J. Chem. Phys.*, 2002, **116**, 8533.
- 40 P. Petrov, S. Miklavic, U. Olsson and H. Wennerström, *Langmuir*, 1995, **11**, 3928.
- 41 M. A. Plunkett, A. Feiler and M. W. Rutland, *Langmuir*, 2003, **19**, 4180.
- 42 A. A. Feiler, L. Bergstrom and M. W. Rutland, *Langmuir*, 2008, **24**, 2274.
- 43 J. E. Sader, J. W. M. Chon and P. Mulvaney, *Rev. Sci. Instrum.*, 1999, **70**, 3967.
- 44 J. Ralston, I. Larson, M. W. Rutland, A. A. Feiler and M. Kleijn, *Pure Appl. Chem.*, 2005, **77**, 2149.
- 45 T. Pettersson, N. Nordgren, M. W. Rutland and A. A. Feiler, *Rev. Sci. Instrum.*, 2007, **78**, 093702.
- 46 M. L. Hair, *J. Non-Cryst. Solids*, 1978, **19**, 299.
- 47 P. M. Claesson, R. G. Horn and R. M. Pashley, *J. Colloid Interface Sci.*, 1984, **100**, 250.
- 48 B. J. Briscoe and D. C. B. Evans, *Proc. R. Soc. London, Ser. A*, 1982, **380**, 389.
- 49 M. He, A. Szuchmacher Blum, G. Overney and R. M. Overney, *Phys. Rev. Lett.*, 2002, **88**, 154302.
- 50 G. Tomlinson, *Phil. Mag.*, 1928, **6**, 695.
- 51 E. Gnecco, R. Bennewitz, T. Gyalog, C. Loppacher, M. Bamberlin, E. Meyer and H. J. Guntherodt, *Phys. Rev. Lett.*, 2000, **84**, 1172.
- 52 T. Bouhacina, J. P. Aimé, S. Gauthier, D. Michel and V. Heroguez, *Phys. Rev. B: Condens. Matter*, 1997, **56**, 7694.